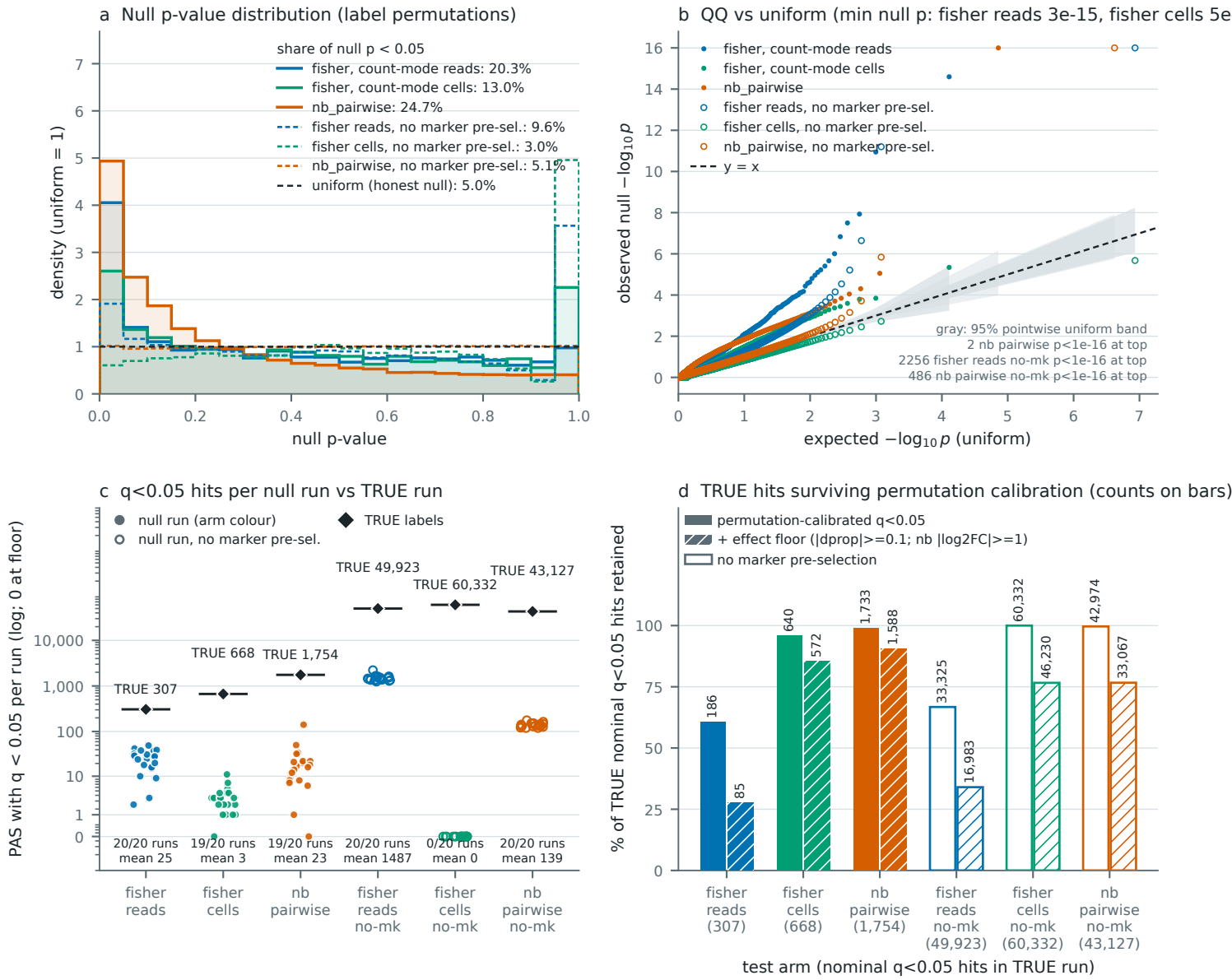


switch diff calibration on a correctly keyed count matrix -- valid FDR control: fisher cells no-mk (conservative); anti-conservative: fisher reads / fisher cells / nb pairwise / fisher reads no-mk / nb pairwise no-mk (testis mouse1 clip-seeded v2, 1,230 STARSolo cells, SPC/RS/ES, 3 stage pairs per run, 20 label permutations)



Input: Stage-1b acceptance run clusters.h5ad (verified keying, layers['counts'], Stage 0c); labels = STARSolo-GEX marker-argmax stage (step4_stage.py, orthogonal to the PAS matrix), restricted to the 1,294 STARSolo cells, SPG dropped (n=64, fragile; manuscript/11) -> 1,230 cells (SPC 370, RS 504, ES 356), 76,711 PAS; 3-stage agreement with PAS-derived labels 98.8%. Null = obs['stage'] permuted across cells (label counts kept; numpy seeds 1-20, the SAME permutations for every arm), full `switch diff` re-run per permutation; main arms include the default top-200 wilcoxon marker pre-selection on .X (label double-dip deliberately inside the null), diagnostic arms (dashed/hollow) switch it off (--marker-top-n 0; fisher reads, no marker pre-sel. 20/20 perms; fisher cells, no marker pre-sel. 20/20 perms; nb_pairwise, no marker pre-sel. 20/20 perms); BH families are per stage pair: fisher, count-mode reads: 20.3% null p<0.05, 7.66% null q<0.05, 20/20 null runs with hits; fisher, count-mode cells: 13.0% null p<0.05, 0.96% null q<0.05, 19/20 null runs with hits; nb_pairwise: 24.7% null p<0.05, 1.27% null q<0.05, 19/20 null runs with hits; fisher reads, no marker pre-sel.: 9.6% null p<0.05, 0.69% null q<0.05, 20/20 null runs with hits; fisher cells, no marker pre-sel.: 3.0% null p<0.05, 0.00% null q<0.05, 0/20 null runs with hits; nb_pairwise, no marker pre-sel.: 5.1% null p<0.05, 0.13% null q<0.05, 20/20 null runs with hits. TRUE vs null hit COUNTS are not directly comparable (permuted labels change the marker set / the >=10-cells filter); per-test null rates are the calibrated quantities. Permutation-calibrated q: empirical p of each TRUE test = rank within the pooled null p-values of the same pair (global-null reference, resolution 1/(N_null+1)), BH per pair; effect floor [delta proportion] >= 0.1 (fisher) or |log2FC| >= 1 (nb). Rule for 'calibrated': calibrated iff null p<0.05 rate <= 7% AND null p<0.01 rate <= 1.5% AND null q<0.05 rate <= 5% AND <= 25% of null runs carry any q<0.05 hit (anti-conservative checks only; KS reported, not gated). Caveats: one mouse, one tissue with very large true stage effects (TRUE hit counts are upper bounds, not precision); label permutation tests exchangeability under the global null of no stage effect -- it does not model within-stage heterogeneity; nb_pairwise dispersion is a per-PAS plug-in clipped to [1e-4, 10] without shrinkage -- nb pairwise: 0.8% of null tests but 13% of null q<0.05 hits sit at the floor; nb pairwise no-mk: 8.5% of null tests but 67% of null q<0.05 hits sit at the floor; fisher p-values are discrete (mass at p=1). Palette six-checks validated 2026-08-19.